

FSM Sandbox: A Mixed-Reality Platform for Error Attribution and Algorithm Validation in Autonomous Driving Systems

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Abstract—Hardware-in-the-Loop (HIL) testing is a critical link for bridging the gap between pure software simulation and real-world deployment in autonomous driving, yet the inherent trade-off between high cost and high fidelity has limited its widespread adoption in academic research. To address this challenge, this paper introduces the FSM Sandbox, a mixed-reality testbed whose core objective is to enable precise localization and attribution of errors within autonomous driving systems. The central technical innovation of this work is a novel hybrid coordinate registration algorithm, which effectively compensates for the intrinsic nonlinear errors of consumer-grade positioning hardware (HTC Vive Tracker) through a three-stage “robust estimation → rigid alignment → non-rigid correction” pipeline, providing the platform with centimeter-level positioning accuracy. More importantly, we propose and apply a HIL testing methodology we term “Diagnostic Decoupling.” This methodology, realized through a specific system architecture, achieves, for the first time, the quantitative separation and attribution of Planning Error and Execution Error within the autonomous driving software stack. By deploying the mainstream Autware system on the platform for a point-to-point navigation task and conducting an in-depth analysis of its multi-level trajectories, we successfully decompose the end-to-end tracking error into its respective sources, providing clear optimization directives for algorithm developers. This research demonstrates that the platform is not merely a high-fidelity validation tool but a powerful diagnostic instrument.

Index Terms—Autonomous Driving, Error Attribution, Hardware-in-the-Loop (HIL), Diagnostic Decoupling, Mixed Reality, Algorithm Validation, CARLA.

I. INTRODUCTION

THE commercial deployment of Autonomous Vehicles (AVs) is at a critical juncture, with the primary challenge evolving from functional implementation to the effective validation of safety across an immense number of “long-tail” scenarios. A seminal study by the RAND Corporation highlighted that demonstrating an AV’s safety superiority over human drivers through physical road testing alone would necessitate billions of kilometers, a requirement that is economically and temporally infeasible [1]. Consequently, building novel testing paradigms that merge the flexibility of virtual testing with the authenticity of the physical world has become a critical and urgent breakthrough area in the field.

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A layered testing ecosystem centered on “X-in-the-Loop” (XiL) methodologies has become an industry-wide consensus. Software-in-the-Loop (SiL) simulations provide a scalable environment for early-stage algorithm iteration, but their utility is constrained by the “sim-to-real” gap [3], [4]. To bridge this gap, Hardware-in-the-Loop (HiL) testing significantly enhances credibility by incorporating real hardware components [5]. However, existing testing paradigms face a **methodological trilemma**, an irreconcilable trade-off among **testing fidelity, cost-effectiveness, and iteration efficiency**.

A deeper challenge lies in the fact that an autonomous driving system is a deeply coupled, complex system, where its end-to-end performance emerges from the nonlinear interactions of perception, localization, planning, and control modules. This inherent coupling leads to an **ambiguity in causal attribution**: when the system exhibits suboptimal behavior during a test, precisely attributing the observed performance issue to its root cause becomes exceedingly difficult due to the simultaneous presence and mutual influence of multiple potential error sources.

To address this challenge, the core objective of this paper is not merely to build a low-cost HIL platform but to propose a testing methodology capable of resolving the aforementioned causal ambiguity. We introduce the **FSM Sandbox**, a mixed-reality testbed designed around a core principle we term “**Diagnostic Decoupling**.” Through innovative software algorithms and system architecture, this platform aims to transform the testbed from a mere “integrator” for macroscopic functional validation into a “diagnostic instrument” for the rigorous, quantitative evaluation of specific algorithm modules.

The main contributions of this paper are summarized as follows:

- **We propose and systematically evaluate a novel hybrid coordinate registration algorithm.** Through comparative experiments with various models, we validate the superiority of our proposed “Affine+RBF” model in compensating for consumer-grade positioning hardware errors, providing a key technical foundation for low-cost, high-precision HIL testing.
- **We propose and implement a HIL testing methodology centered on the principle of “Diagnostic Decoupling.”** This methodology, through a specific system architecture, achieves the quantitative separation of planning and execution errors in autonomous driving systems.
- **We demonstrate the diagnostic capabilities of our platform through a case study on the Autware system.** We

TABLE I
COMPARATIVE ANALYSIS OF ADVANCED MINIATURE AUTONOMOUS DRIVING TEST PLATFORMS

Feature Dimension	FSM Sandbox (Ours)	ICAT [10]	μ City [11]
Core Goal	Error Attribution & Decoupled Validation	Vehicle-in-the-Loop System	Mixed-Reality Interaction
Positioning Scheme	External Ground Truth (Vive)	Onboard SLAM (2D LiDAR)	External Ground Truth
Error Coupling	Diagnostic Decoupling	Coupled (System Demonstrator)	Decoupled
Digital Twin Sync	Real-time Bidirectional	Real-time Bidirectional	Real-time, primarily R2V
Virtual Environment	High-Fidelity (CARLA)	High-Fidelity (CARLA)	Configurable Virtual Elements
Cost & Agility	Very Low Cost, High Agility	Medium Cost, Medium Agility	High Cost, Medium Agility

successfully decompose the end-to-end tracking error into its respective sources, providing clear, actionable insights for algorithm optimization and proving the unique value of the platform as an error attribution and algorithm validation tool.

II. RELATED WORK

The challenges in AV validation have spurred research into miniaturized test platforms, seeking a balance between pure simulation and full-scale testing. While platforms like Duckietown [6], [8] and F1TENTH [7], [9] have made significant contributions to education and rapid algorithm prototyping, their primary objective is not to provide a system-level, high-fidelity validation environment.

More directly related to our research are platforms that employ digital twin technology, with ICAT [10] and μ City [11] being representative state-of-the-art systems. To systematically elucidate the novelty of our work, we present a multi-dimensional comparison in Table I.

As shown in Table I, the **architectural choice of the positioning subsystem is the fundamental differentiator of our work**. ICAT, for instance, relies on onboard 2D LiDAR SLAM, which introduces a critical **methodological confound**: the performance of the positioning system is **coupled** with the performance of the planning and control algorithms under test. This ambiguity limits its effectiveness as a pure algorithm validation platform.

In contrast, the FSM Sandbox is architecturally designed to eliminate this ambiguity. **The core technical contribution of our research is the development of an advanced hybrid registration algorithm that empowers a low-cost external positioning system to provide high-precision ground truth, upon which a "Diagnostic Decoupling" test framework is built**. It transforms the platform from a system demonstrator into a **pure, reliable, and quantifiable algorithm diagnostic testbed**, where the System Under Test (SUT) can be evaluated in an environment free from confounding variables, and error sources can be attributed.

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION OF DIAGNOSTIC DECOUPLING

To translate the "Diagnostic Decoupling" principle from a theoretical concept into an operational engineering entity, we designed and implemented the FSM Sandbox system architecture. This section details the three core pillars of this architecture: first, the digital twin data framework, which forms the system's "brain" and is centered on diagnostic

decoupling; second, the positioning and registration subsystem that provides high-precision data input to this framework, serving as the system's "eyes"; and finally, the physical world subsystem that hosts all testing activities, acting as the system's "body." The overall architecture, depicted in Fig. 1, constitutes a complete "perception-planning-control" closed loop.

A. Digital Twin Framework with Diagnostic Decoupling

To address the challenge of causal ambiguity arising from the deep coupling of subsystems in autonomous driving, we propose and implement a digital twin framework based on the core design principle of **Architectural Decoupling**. We define this design philosophy, aimed at enhancing causal attribution by controlling variables, as **"Diagnostic Decoupling."** The core objective is to transform the testbed from an "integrator" for macroscopic functional validation into a "diagnostic instrument" for the rigorous, quantitative evaluation of specific algorithm modules. This principle is realized through two architecturally designed, asymmetric data streams: the **Real-to-Virtual (R2V)** and **Virtual-to-Real (V2R)** data links.

1) *R2V Data Link: Foundation for Decoupling Localization and Perception:* The R2V data link is the cornerstone of our framework's diagnostic decoupling capability. It systematically decouples localization error at the architectural level through **Forced State Synchronization**. This mechanism utilizes an independent, high-precision external positioning system (see Section 3.2) to acquire the ground-truth pose of the physical vehicle. A dedicated ROS 2 node then frequently calls the CARLA simulator's `vehicle.set_transform()` API to forcibly synchronize this pose to the virtual twin. This operation intentionally bypasses CARLA's own physics engine for the ego vehicle, replacing an unknown and coupled onboard localization error source with a known and independent external reference. This allows the subsequent planning and control stack to operate on a deterministic, high-precision pose baseline.

Furthermore, by providing the simulator with a ground-truth pose, the R2V link indirectly facilitates the **decoupling of perception error**. Once the virtual twin's position and orientation are precisely anchored to the physical world's ground truth, CARLA can render idealized and deterministically reproducible sensor data based on this pose. This step is methodologically crucial as it substitutes noisy and stochastic physical sensor data with clean, unbiased virtual perception data.

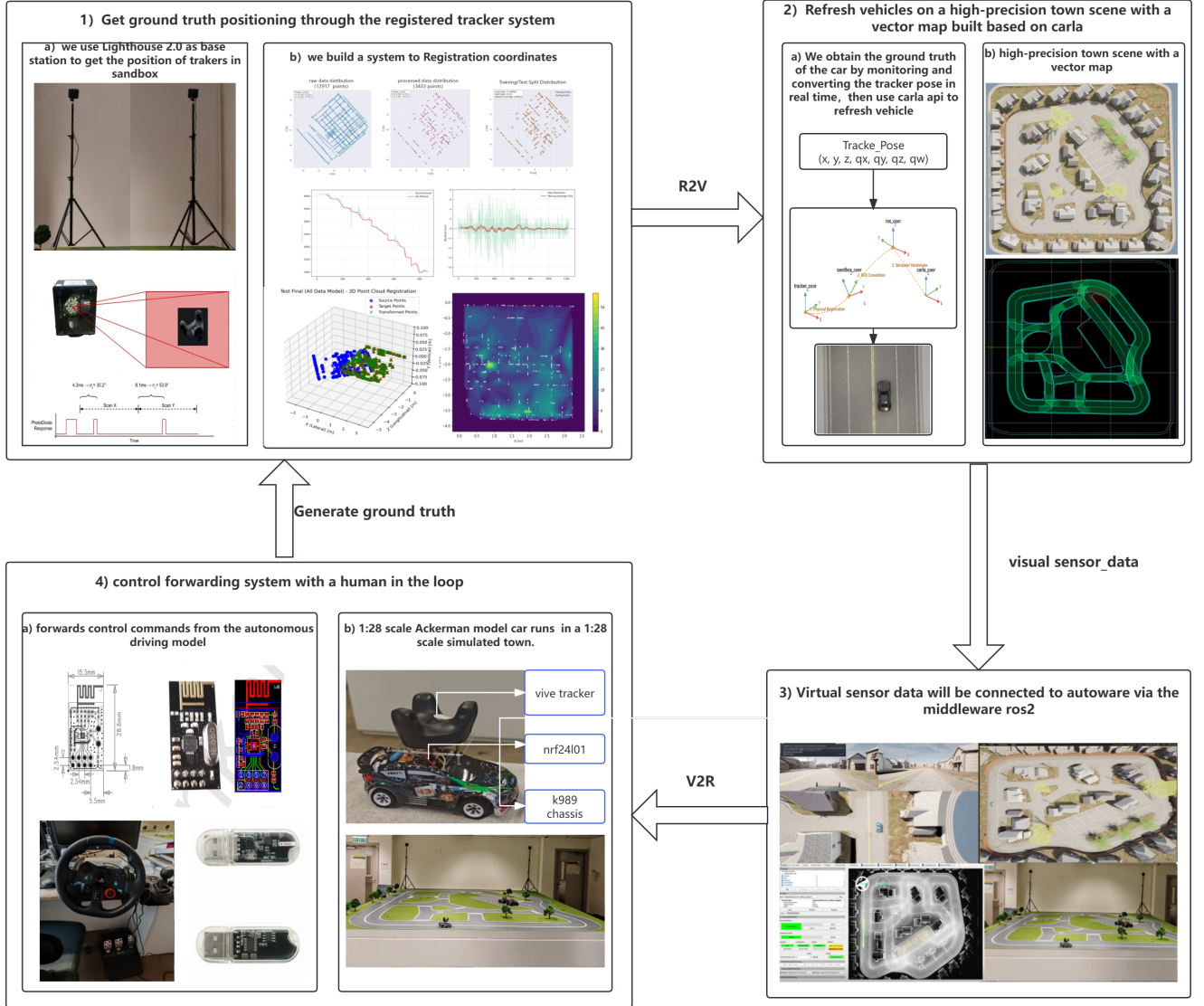


Fig. 1. The overall architecture of the FSM Sandbox. The system seamlessly integrates a physical miniature vehicle with its digital twin in the CARLA simulator. A Real-to-Virtual (R2V) data link synchronizes the ground-truth physical state to the virtual world, where an autonomous driving stack (e.g., Autoware) generates control commands based on virtual sensor data. A Virtual-to-Real (V2R) link then transmits these commands back to the physical vehicle for execution, forming a complete, closed-loop HIL testbed designed for architectural decoupling.

2) V2R Data Link: Closing the Physical Control Loop with Decoupled Decisions: While the R2V link serves as the “input” for decoupling, the V2R link is the “output” for execution and validation. Its core function is to close a physical control loop based on the decoupled inputs. In this architecture, the AV’s planning module operates in the virtual environment, with its decisions based entirely on the aforementioned “pure” information. The control commands generated by the planner (e.g., desired velocity and steering angle) are then transmitted via the V2R link from the simulation environment to the physical vehicle’s low-level controller for execution.

The methodological value of the V2R link lies in its ability to translate a decision made in a controlled, idealized information space into observable and measurable vehicle behavior in the physical world. It enables researchers to

evaluate the ultimate performance of an algorithm’s core logic when interacting with real-world physics (e.g., tire slip, actuator latency), having isolated it from the uncertainties of localization and perception. It is noteworthy that the entire closed loop from R2V perception to V2R execution has an inherent system latency. We measured the mean end-to-end latency of our system to be approximately 65 ms, a component of the “Execution Error” discussed in our analysis.

In summary, by establishing a foundation of dual decoupling through the R2V link and closing the physical validation loop with the V2R link, our framework systematically decouples localization and perception errors at an architectural level, decomposing a highly coupled system validation problem into a series of nearly orthogonal, independently analyzable subproblems.

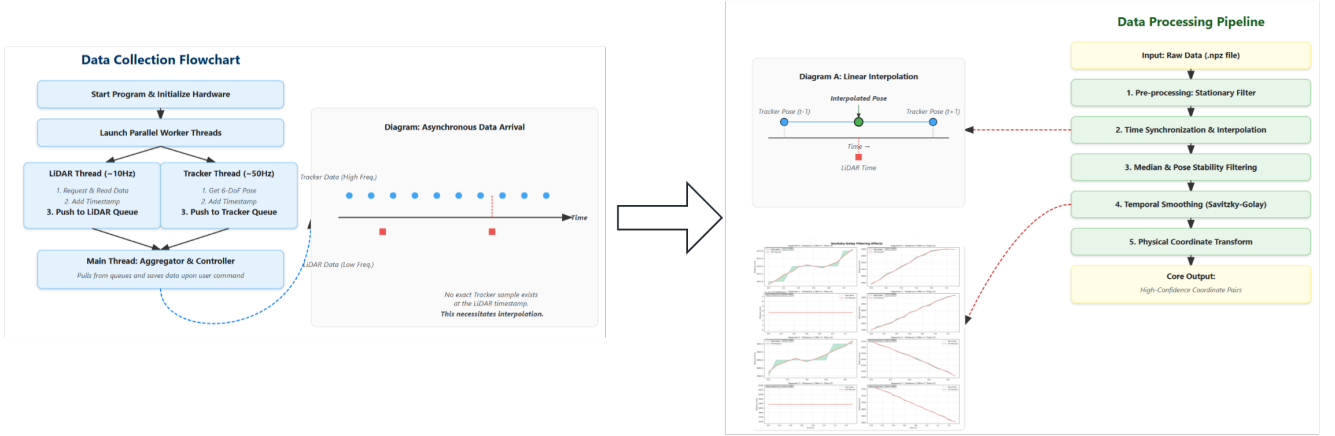


Fig. 2. The data processing and validation pipeline for coordinate registration. This multi-stage process refines raw sensor readings into high-confidence coordinate pairs. Key stages include: (1) Stationary filtering to remove redundant data; (2) Time synchronization to align LiDAR and Tracker data; (3) Median and pose stability filtering to remove outliers; (4) Temporal smoothing with a Savitzky-Golay filter; and (5) Transformation into physical world coordinates.

B. High-Precision Positioning and Hybrid Coordinate Registration

A positioning subsystem capable of providing centimeter-level, high-refresh-rate ground truth is an indispensable prerequisite for the "Forced State Synchronization" described in Section 3.1. To achieve this under strict cost constraints, we devised a solution based on consumer-grade hardware and advanced software compensation algorithms.

1) *Hardware Configuration and Data Acquisition:* Our positioning solution employs an HTC Vive Tracker 3.0 with two Base Station 2.0 units. This configuration provides centimeter-level accuracy at a cost significantly lower than professional motion capture systems (e.g., Vicon). To acquire physical-world coordinates for registration, we designed a **calibration tool** integrating the Vive Tracker with two orthogonally mounted single-point LiDARs, used to measure perpendicular distances to two physical reference planes (e.g., walls). During data acquisition, this tool is moved freely within the operational area. A custom ROS2 node developed using the Python 'openvr' library records the Tracker's 6-DoF pose at 50Hz, while another node synchronously records the two LiDAR distance readings. These time-stamped data streams are saved as a raw dataset for subsequent offline processing.

2) *Multi-stage Data Processing and Validation Pipeline:* The accuracy of any registration algorithm is fundamentally limited by the quality of its input data. To address this, we designed and implemented a **multi-stage data processing and validation pipeline** centered on a physical model to **distill** high-confidence coordinate pairs from noisy raw measurements. This pipeline, visualized in Fig. 2, includes state-machine-based redundant data filtering, strict time synchronization and pose stability validation, and physics-model-based time-domain filtering and correction. Through this pipeline, raw sensor readings are systematically transformed into a high-confidence set of coordinate points 'P_phy_i' in the physical world frame.

3) *Hybrid Coordinate Registration Algorithm: A Systematic Model Exploration:* To achieve precise alignment between the physical and virtual coordinate systems, we propose a **hybrid coordinate registration method**. We decouple the treatment of rotation and translation components of the pose. For rotation, we use the standard **SVD-based absolute orientation method**. For position, we designed and implemented a **three-stage position registration pipeline**:

- 1) **Robust Outlier Rejection:** We use RANSAC to identify and reject outliers from the dataset.
- 2) **Rigid Coarse Registration:** On the cleaned inlier set, we use SVD to find the optimal rigid transformation that establishes a baseline linear alignment.
- 3) **Non-Rigid Deformation Modeling and Comparison:** To correct for the spatially correlated nonlinear distortions inherent in consumer-grade positioning systems, we systematically evaluated several non-rigid deformation models, including **Affine Transformation**, **Radial Basis Function (RBF)**, **Gaussian Process (GP)**, and **Multi-Layer Perceptron (MLP)**. Our experiments (detailed in Section 4.1) demonstrate that a hybrid model combining an affine transformation with an DeepMLP interpolator achieves the best registration accuracy.

C. Physical World Subsystem

The physical world subsystem serves as the final execution and interaction medium for the entire HIL testing loop. It consists of the miniature vehicle, the physical sandbox, and a human-machine interface.

- **Miniature Vehicle:** We designed and built the core physical actuation unit based on a 1:28 scale RC car chassis. The vehicle is intended to accurately reproduce the **first-order kinematic properties** of a real vehicle. Its retention of **Ackermann Steering Kinematics** is crucial for simulating the turning characteristics of real vehicles at low speeds. The model intentionally simplifies complex tire-ground dynamics to focus the analysis on

TABLE II
COMPARATIVE PERFORMANCE METRICS OF REGISTRATION METHODS ON THE TEST SET (ERROR IN MM)

Registration Method	Test RMSE (mm)	Test MAE (mm)	Train RMSE (mm)	Max Error (mm)	95th Percentile (mm)
SVD Rigid Transform	16.25	14.28	16.63	43.3	30.1
RBF Registration	12.29	7.96	12.63	57.2	30.3
CPD Registration	14.64	11.57	15.07	41.8	29.9
Hybrid (Affine + MLP)	10.18	7.63	10.00	35.0	21.7
Hybrid (Affine + Ensemble)	9.30	6.87	10.52	36.1	19.2
Adaptive Quadtree Hybrid	55.56	36.97	55.90	132.4	108.3

the geometric path-tracking capabilities of planning and control algorithms.

- **1:28 Town Physical Sandbox:** We constructed a 1:28 scale physical town model to provide realistic reflections and occlusions for the virtual LiDAR and cameras, enhancing the immersion and physical realism of the mixed-reality tests.
- **Human-Machine Interface (Logitech G29):** To support **Human-in-the-Loop (HiL)** testing, we integrated a Logitech G29 force-feedback steering wheel, which can be used for efficient expert data collection or for real-time intervention and takeover during tests.

D. Core Technology Validation: Positioning Algorithm Comparison

To establish a high-precision ground truth, which is the bedrock of our "Diagnostic Decoupling" methodology, we must first validate and select the optimal algorithm for compensating the inherent non-linear errors of the consumer-grade HTC Vive positioning system.

1) *Experimental Setup:* We collected a dataset of over 10,000 coordinate pairs using the calibration tool described in Section 3.2. This dataset maps the Vive Tracker's reported coordinates (source) to the high-precision LiDAR-measured coordinates (target). The data was split into a training set (80%) and a test set (20%) to ensure a robust evaluation of model generalization. Before comparing the registration models, it is crucial to visualize the effectiveness of the data processing pipeline. As shown in Fig. 3, the pipeline provides a dramatic improvement in data quality by removing 73.8% of the initial points, which are identified as noisy or redundant. This ensures that the subsequent algorithm comparison is performed on a high-confidence dataset.

We compared a range of registration methods, from simple rigid transforms to complex non-rigid and hybrid models: SVD Rigid Transform, RBF Registration, CPD Registration, two hybrid models combining affine transformation with machine learning correctors (MLP and an Ensemble model), and an Adaptive Quadtree Hybrid model.

2) *Results and Analysis:* The performance of all evaluated methods on the test set is summarized in Fig. 4 and Table II. The results yield several key insights:

- **Insufficiency of Rigid Transformation:** The baseline SVD Rigid Transform yields a Test RMSE of 16.25 mm. While a significant improvement over unaligned data, this relatively high error confirms the presence of

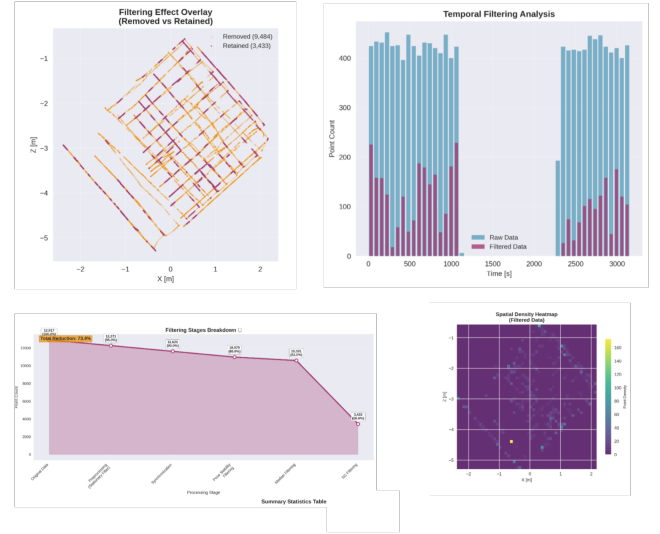


Fig. 3. Visual analysis of the data processing pipeline's effectiveness. (Top-Left) Spatial overlay of data points removed during filtering (purple) versus those retained (orange). (Top-Right) Temporal distribution of points before (blue) and after (purple) filtering, showing significant reduction in redundant data. (Bottom-Left) Point count reduction at each stage of the pipeline, illustrating a 73.8% total reduction. (Bottom-Right) Spatial density heatmap of the final, cleaned data set used for registration.

substantial non-linear, spatially-varying distortions in the Vive system that a rigid model cannot capture.

- **Effectiveness of Non-Rigid Correction:** All non-rigid and hybrid methods outperform the rigid SVD model. This underscores the necessity of a non-rigid correction stage. The RBF Registration (12.29 mm Test RMSE) and our Hybrid (Affine + Ensemble) model (9.30 mm Test RMSE) show particularly strong performance, reducing the error by 24% and 43% respectively compared to SVD.
- **Superiority of the Hybrid Approach:** The two hybrid models, which first perform a global affine alignment and then train a machine learning model on the residuals, yield the best results. The **Hybrid (Affine + Ensemble)** model emerges as the clear winner, achieving the lowest Test RMSE (9.30 mm) and Test MAE (6.87 mm). This demonstrates the effectiveness of a staged "robust estimation → non-rigid correction" pipeline.
- **Analysis of Model Failure:** Interestingly, the sophisticated Adaptive Quadtree Hybrid model performed worst, with a Test RMSE of 55.56 mm. A detailed look at its error distribution reveals that while it corrects many

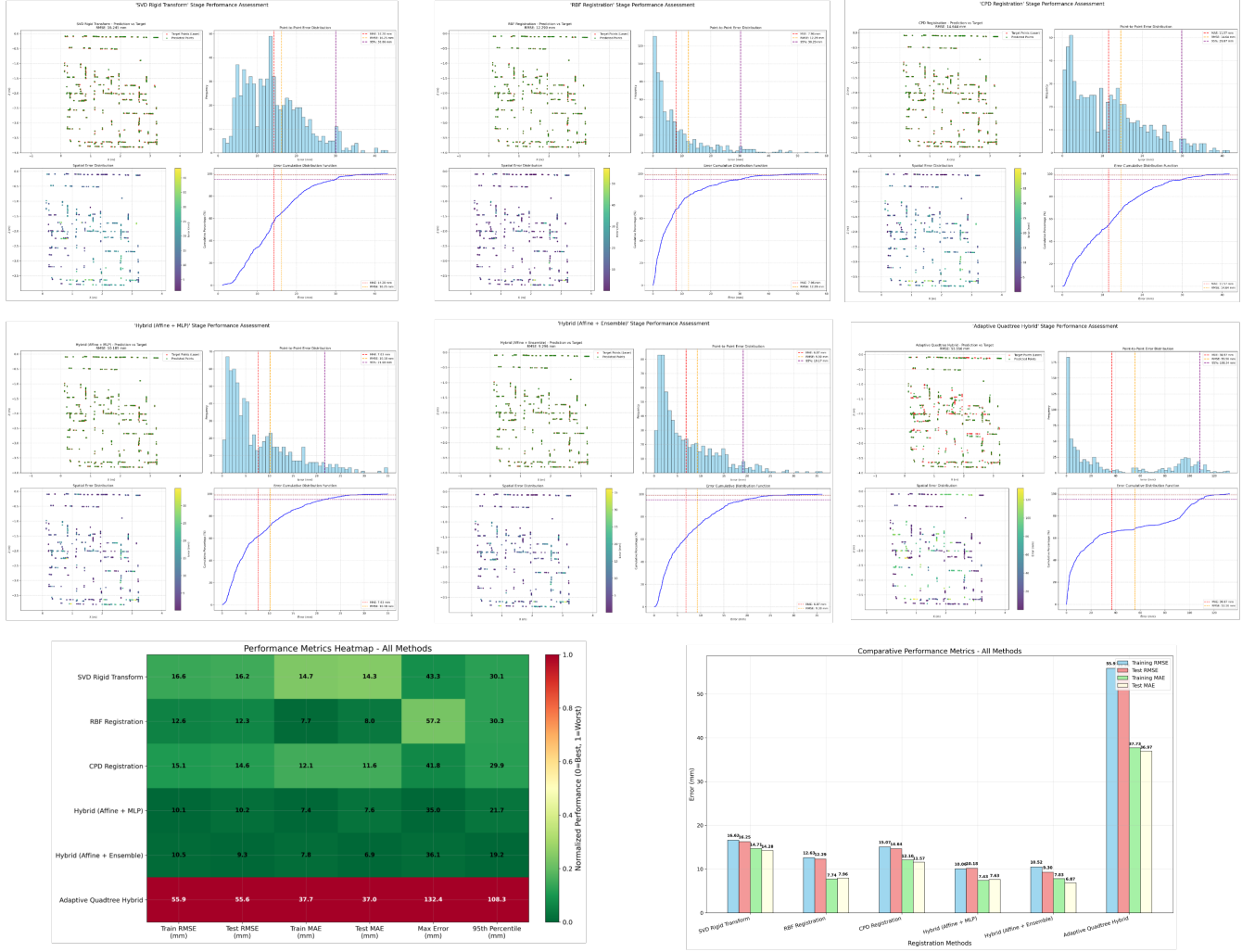


Fig. 4. Comparative performance of all registration methods. The bar chart clearly shows the error metrics (in mm) for both training and test sets. The hybrid models, particularly the "Hybrid (Affine + Ensemble)" method, demonstrate superior performance with the lowest Test RMSE and MAE, indicating the best generalization. The poor performance of the Adaptive Quadtree model highlights the risk of model mismatch.

points accurately, it introduces very large errors for a subset of points, likely due to overfitting in sparsely populated quadtree cells or instability at cell boundaries. This highlights the importance of choosing a model that is robust and well-suited to the specific error characteristics of the system.

Based on this comprehensive analysis, the **Hybrid (Affine + Ensemble)** model was selected and integrated into the FSM Sandbox. Its sub-centimeter accuracy provides the reliable ground truth required for the subsequent error attribution analysis.

E. Integrated System Performance: Trajectory Tracking and Error Attribution Analysis

After validating the precision of the positioning subsystem, we assessed the diagnostic capability of the integrated framework. We deployed Autoware.Universe as the System Under Test (SUT) to perform a point-to-point navigation task. The goal was not merely to measure performance, but to use

our "Diagnostic Decoupling" architecture to attribute tracking errors to their sources. To achieve this, we analyze three key data streams:

- 1) **Global Path:** The high-level, pre-defined reference route.
- 2) **Local Trajectory:** The fine-grained, feasible path generated in real-time by Autoware's local planner.
- 3) **Actual Vehicle Position:** The high-precision ground truth of the physical vehicle, captured by our validated positioning system.

From these, we define two primary error components:

- **Planning Error:** The Cross-Track Error (CTE) between the *Local Trajectory* and the *Global Path*. This isolates the performance of the motion planning algorithm.
- **Execution Error:** The CTE between the *Actual Vehicle Position* and the *Local Trajectory*. This quantifies the combined error from the control commands, V2R data link latency, and physical vehicle dynamics.

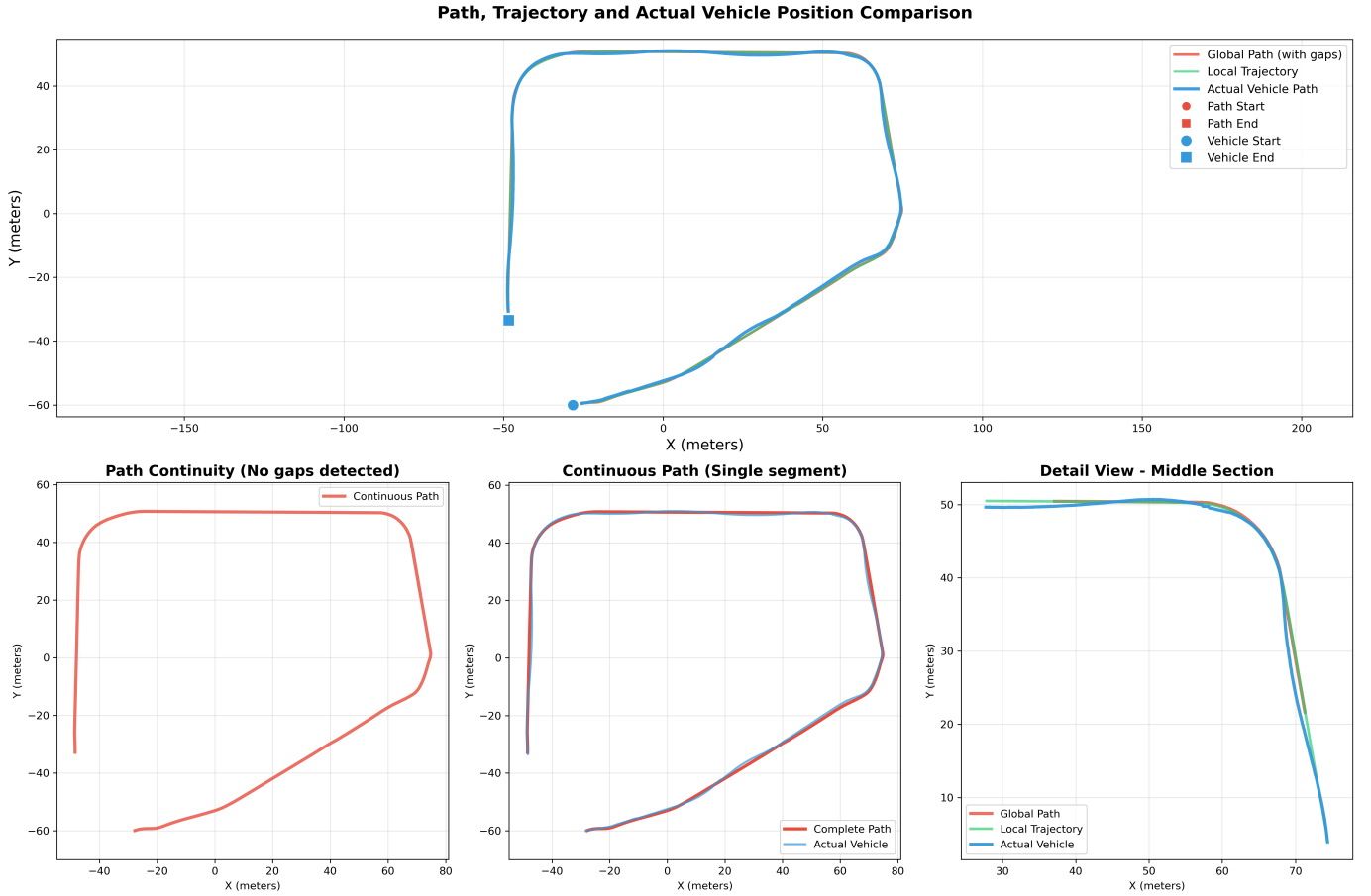


Fig. 5. Qualitative overview of the Autware navigation test. The main plot shows the Global Path (red), the Local Trajectory generated by the planner (green), and the Actual Vehicle Path (blue). While the vehicle follows the general route, visible deviations motivate a deeper quantitative analysis. The subplots provide detailed views of path continuity and a magnified section, showing the close but imperfect tracking.

Fig. 5 provides a qualitative overview of the test run, showing that the vehicle successfully navigates the course. However, the true diagnostic power of the FSM Sandbox is revealed through the quantitative analysis in Fig. 6.

- **Error Attribution in Action:** By comparing the top and middle plots of Fig. 6, we can directly attribute error. For instance, at approximately 08:33:00, the total error (top plot, blue line) spikes to over 1.0 meter. In the same time frame, the execution error (middle plot, green line) remains low and stable, around 0.2 meters. The significant difference between these two values is the **Planning Error**. This provides a clear, quantitative diagnosis: in this corner, the primary performance bottleneck was Autware’s local planner struggling to follow the sharp curve of the global path, not the vehicle’s physical inability to execute commands. This gives developers a precise target for optimization—tuning the planner’s parameters for cornering, rather than the vehicle’s low-level controller.
- **Identification of Systematic Bias:** The bottom plot of Fig. 6 reveals another critical insight. The mean CTE values for all methods are consistently positive (e.g., the V2P mean is +0.314m). This indicates a **systematic bias**: the vehicle consistently drives to the right of its

reference path. This type of error is unlikely to be caused by the planner’s logic; instead, it points directly to a physical calibration issue, such as a steering angle offset, an incorrect sensor mounting transformation, or a bias in the low-level motor controller.

This case study successfully demonstrates that the FSM Sandbox is more than a validation platform; it is a diagnostic instrument. By architecturally decoupling error sources, it provides clear, actionable insights, enabling developers to distinguish between algorithmic weaknesses (planning error) and system integration issues (execution error and systematic bias), thus accelerating the development and refinement cycle.

IV. DISCUSSION AND LIMITATIONS

The core advantage of the FSM Sandbox proposed in this paper lies in its ability to achieve quantitative attribution of autonomous driving system errors through innovative algorithms and architecture. However, we must also clearly recognize its limitations and how they influence the interpretation of experimental results.

- **Impact of Limitations on Result Interpretation:**
 - **Dynamics Gap:** This platform simulates the vehicle’s **kinematics**, not its complex **dynamics**. This

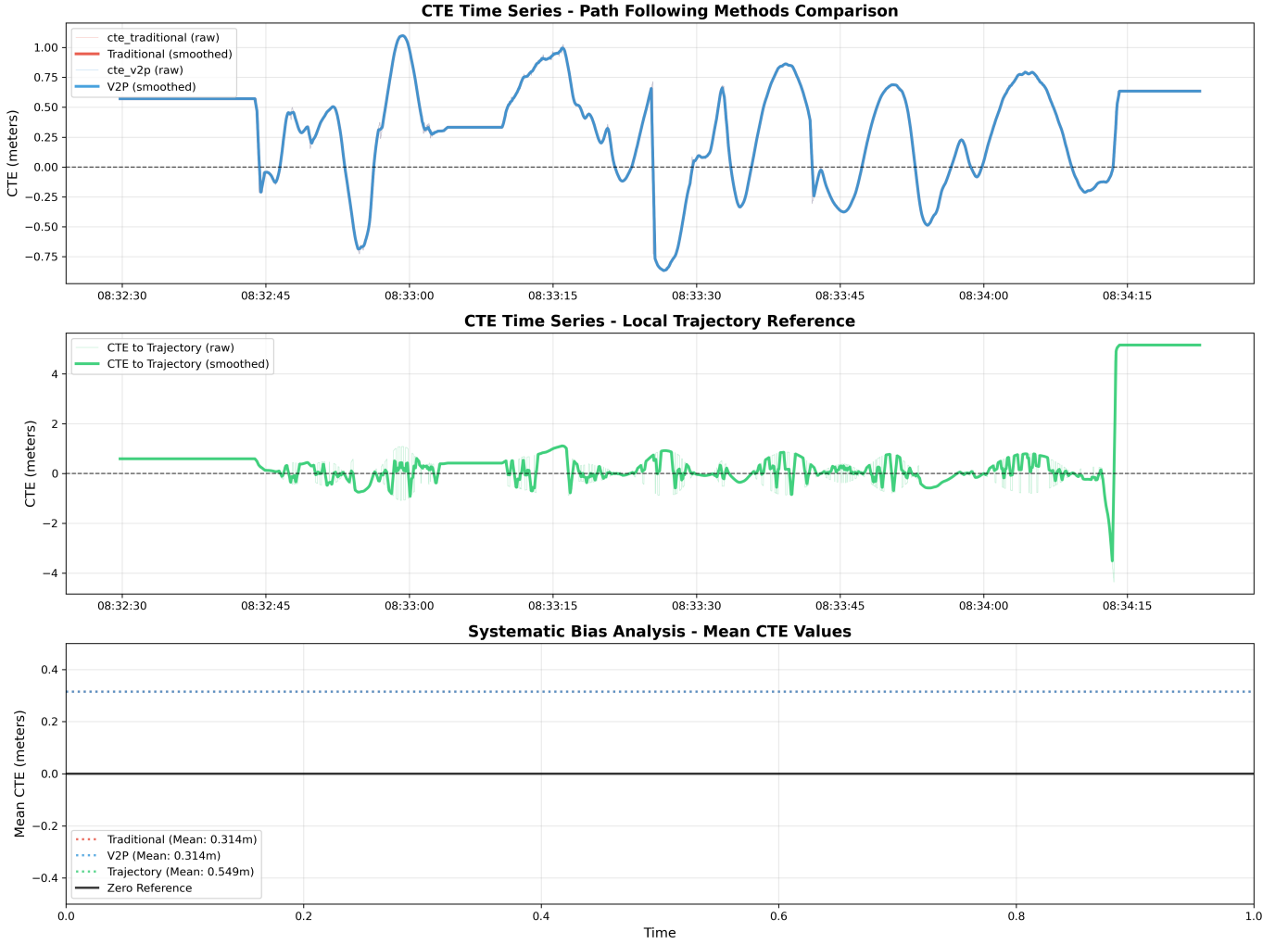


Fig. 6. Quantitative error attribution analysis via CTE time series. **(Top)** The total end-to-end tracking error (vehicle vs. global path), showing significant peaks. **(Middle)** The execution error (vehicle vs. local trajectory), which is generally smaller and more stable. **(Bottom)** Systematic bias analysis, revealing a consistent positive mean CTE across different calculation methods, indicating a vehicle offset.

limitation means that the **Execution Error** we observed in Section 4.2 primarily reflects data link latency and motor control performance, without capturing the challenges to controller stability posed by dynamic effects like body roll and pitch during high-speed cornering in a real vehicle. Therefore, while our platform can effectively evaluate the geometric path generation capability of planning algorithms, its ability to validate controller stability in high-speed dynamic scenarios is limited.

- **Perception Gap:** By employing idealized virtual perception, the platform cannot evaluate the impact of real-world sensor noise on the stability of Autoware’s planning and control modules. The **Planning Error** we observed was measured under a ‘best-case’ scenario of perfect perception, providing a clear upper-bound benchmark for the algorithm’s geometric tracking performance.

- **Scalability:** The current research focuses on single-vehicle validation scenarios. The centralized positioning

solution based on HTC Vive may face challenges with wireless signal interference and increased computational load when scaled to multi-agent systems. Extending this framework to large-scale multi-vehicle collaborative scenarios is an important direction for our future work.

V. CONCLUSION

This paper addressed the problem of causal ambiguity in the autonomous driving validation domain by proposing the FSM Sandbox—a mixed-reality platform for error attribution and algorithm validation. The core contribution of our work lies not only in implementing a low-cost HIL platform but, more importantly, in proposing and validating a “**Diagnostic Decoupling**” testing methodology. We first developed an “Affine+RBF” hybrid registration algorithm through systematic comparative experiments, providing a foundation of centimeter-level positioning accuracy for the platform. Subsequently, through an error attribution analysis of the Autoware system, we demonstrated that this methodology can decompose complex end-to-end system performance issues

into quantifiable subproblems attributable to specific modules (planning or control). This ability to provide diagnostic insights offers an effective and economical path to accelerate the iteration and optimization of autonomous driving algorithms. This research provides the academic community with a fully validated, diagnosis-centric HIL testing solution, contributing to the rapid and efficient iteration and validation of autonomous driving algorithms.

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